Redundancy and Geography in Supply-Network Cascades:
Extending the Elliott–Golub–Leduc Model to k-of-n Sourcing and Regional Shocks

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Abstract

Elliott, Golub, and Leduc [1] (EGL) show that supply networks can collapse suddenly, Carvalho, Nirei, Saito, and Tahbaz-Salehi [2] (CNST) measure how the 2011 Japanese earthquake spread through real supply links, and Dorogovtsev, Goltsev, and Mendes [3] (DGM) explain why k-cores of random graphs vanish abruptly. We extend EGL's model in two ways: a firm needs at least k working suppliers for each input, and a regional disaster shuts down every firm in a region at once. We show that collapse becomes sudden as soon as a firm needs two of anything, and that needing two working suppliers instead of one cuts the tolerable regional shock more than fivefold (from 42% to 8% of regions hit in our base case). With one input, spreading suppliers out or co-locating them gives the same breaking point, and spreading never hurts when one supplier is enough. When firms need two suppliers, however, splitting them into two regional pairs can make the network break at a smaller shock than either extreme (17.1% versus 19.5% with one input). Simulations of networks with 420,000 firms match the thresholds predicted for 20 layers to within 0.001.

1 Introduction

Supply networks are a natural setting for the cascades studied in this course: a firm missing one essential input cannot produce, and its customers then face the same problem. Firms protect themselves in two ways: they buy each input from several suppliers (*redundancy*), and they place those suppliers in different regions (*geography*). We ask how these two protections move the point at which a supply network collapses.

We read three papers, none linked from the course page. EGL is our starting point: it models supply networks as random trees and finds a "precipice" in output. CNST is the leading empirical study of supply-chain contagion, and the shock it studies is regional. DGM analyzes k-cores, which collapse abruptly for the same mathematical reason as our model. Sections 2–5 discuss and connect the papers, Sections 6–7 propose and test our model, and Section 8 outlines next steps.

2 Paper 1: Elliott, Golub, and Leduc (2022), "Supply Network Formation and Fragility"

Technical content. Firms sit in layers of a deep random supply tree. A firm needs m essential inputs and has n potential suppliers for each in the layer below; each relationship works independently with probability x (*relationship strength*). A firm is functional if every input has at least one working relationship with a functional supplier. If each supplier is functional with probability r, a firm is functional with probability

$$R_x(r) = \left(1 - (1 - xr)^n\right)^m, \tag{1}$$

and in a deep network the *reliability* (the probability that a firm works) is the largest fixed point of R_x. For $m \geq 2$, R_x is convex then concave, so as x falls, the largest crossing with the diagonal disappears at a tangency, and reliability *jumps* to zero at a critical strength x_{crit} (their Proposition 1). For $m = 2$ and $n = 4$ we compute $x_{\text{crit}} = 0.518$, where reliability drops from 0.64 to 0. For $m = 1$ the drop is continuous and happens at $x = 1/n$. EGL then let firms pay to raise their own x. Firms ignore how their reliability helps their customers, so for $m \geq 3$ a whole range of cost parameters puts the equilibrium just above x_{crit} (their Theorem 1), where a tiny drop in baseline strength destroys production.

Relevance to the course. The failure rule is a threshold rule like the adoption rule of Granovetter [8] and Easley–Kleinberg [7, §19.2], run in reverse: a firm drops out when, for some input, none of its suppliers remains. With $m = 1$ the recursion is the survival probability of a branching process [7, §21.8]. With $m \geq 2$ it resembles a complex contagion [6]. The new element, in the spirit of [10], is that firms' own choices set how close the network sits to its threshold.

Weaknesses and improvements. (i) Relationship failures are independent, and the only aggregate shock is a uniform drop in x. Real disasters are regional, so co-located suppliers fail together. (ii) One working supplier always suffices, which ignores capacity limits. (iii) The tree has no hubs or cycles, the equilibrium theorem needs $m \geq 3$, and x_{crit} cannot be observed in data. Our model addresses (i) and (ii), and we test supplier hubs in simulation.

3 Paper 2: Carvalho, Nirei, Saito, and Tahbaz-Salehi (2021), "Supply Chain Disruptions"

Technical content. The 2011 earthquake and tsunami hit four prefectures holding 4.6% of Japan's output, whose own losses directly account for only about 0.1 percentage points (pp) of national growth. Using supplier–customer links among 750,237 firms (about 5 suppliers each, heavy-tailed degrees) and a difference-in-differences design, CNST find that sales growth fell by 3.8pp for firms with suppliers in the disaster area and 3.1pp for firms with customers there, shrinking with network distance and robust to a placebo network that randomly rewires links. Their calibrated model then attributes 0.47pp of lost GDP growth to the disaster, versus 0.21pp had links out of the region been cut.

Relevance, weaknesses, and improvements. This is a clean causal measurement of contagion on a real network, and the effect fading with each hop is consistent with smooth, damped propagation. Their model spreads a shock through a damped sum over all supply paths, much like the walk sums behind PageRank [7, §14.6]. Three weaknesses limit what it can say about collapse. First, production in the model substitutes smoothly between inputs and firms never shut down, so output falls smoothly with the shock and cannot show a threshold. Second, exposure is measured as network distance to the disaster area, which treats a firm with 1 of 10 suppliers there the same as one with 10 of 10. Third, the sample drops firms that exited, and exit is the outcome most telling of collapse. Section 8 proposes a test that addresses all three.

4 Paper 3: Dorogovtsev, Goltsev, and Mendes (2006), "k-Core Organization of Complex Networks"

Technical content. The k-core of a graph is what remains after repeatedly deleting every vertex with fewer than k neighbors. DGM compute its size in random graphs with a given degree distribution (the configuration model) after a random fraction $Q = 1 - p$ of vertices is removed. Because these graphs look locally like trees, this reduces to a fixed-point equation for the probability u that an edge leads to a vertex that can stay in the core; for an Erdős–Rényi (ER) graph with mean degree z, $u = p\Pr[\text{Poisson}(zu) \geq k-1]$. For $k = 2$ the core shrinks continuously, like the giant component; for $k \geq 3$ it vanishes with a *jump*. Just before collapse the core is held together by vertices with exactly k core neighbors, so one more deletion sets off a large avalanche of pruning. Solving their equations for $z = 10$, we find the 7-core vanishes once 1.25% of vertices are removed, while the giant component survives until 90% are gone. On simulated graphs with 100,000 vertices, our own pruning code finds every k-core ($k = 2, \ldots, 7$) vanishing within about 0.01 of DGM's thresholds.

Relevance, weaknesses, and improvements. The analysis uses the configuration model and branching arguments from the Random Graphs part of the course, and pruning a core is itself a threshold cascade. But it is mean-field (ignoring short cycles and degree correlations), damage is random and independent, every vertex has the same threshold, and links are undirected. A supply-network version needs directed links, firm-specific thresholds, and correlated damage; our model adds the first and last.

5 Synthesis and Research Questions

EGL and DGM both find that a network collapses suddenly, rather than gradually, when each node needs more than one thing to keep working: EGL's firms need several inputs, and DGM's core vertices need several neighbors. CNST, by contrast, see smooth propagation after a large regional shock. The views are compatible: CNST's model cannot produce a jump, their sample drops firms that exited, and in EGL's model a network far above its tipping point also absorbs shocks smoothly.

Two features of real supply chains are missing from all three papers. First, a firm often needs more than one working supplier for an input, because of capacity limits or dual-sourcing policies. Second, suppliers in the same region fail together. EGL and DGM assume independent failures, and CNST's model has no backup suppliers. We therefore ask: **Q1:** If a firm needs $k \geq 2$ working suppliers for an input, does collapse become sudden even when firms have only one input? **Q2:** Is it safer to spread a firm's suppliers across regions or to keep them together, and does the

answer depend on k? **Q3:** When a region is hit, does the number of firms it holds or the amount of supply it provides matter more for collapse?

Closest prior work. Papachristou, Rahimian, and Azadegan [11] also study supply-chain cascades with percolation, but one surviving supplier suffices and regional risk enters only as a failure correlation, which they find lowers resilience, so they recommend geographic diversification. In three-tier chains, Ang, Iancu, and Swinney [4] show that shared tier-2 suppliers make tier-1 disruptions coincide, and Bimpikis, Fearing, and Tahbaz-Salehi [5] that, with S-shaped production, upstream dual sourcing can hurt the manufacturer. We model regions explicitly in a full cascade, solve the two extremes in closed form (Claim 2), and note that needing $k \geq 2$ suppliers alone makes coverage S-shaped; by a mechanism like that of [5], this gives, to our knowledge, the first network-wide case in which regional pairs are worst (Example 1).

6 Our Model: k-of-n Sourcing with Regional Shocks

Setup. We keep EGL's layered tree. Each firm needs m inputs, has n potential suppliers for each in the layer below, and each relationship works with probability x. We make two changes.

- **k-of-n sourcing.** An input is covered only if at least k of its n suppliers are *usable*, meaning the supplier works and so does the relationship. EGL is the case $k = 1$.
- **Regional shocks.** Firms are spread over many regions. A disaster hits each region independently with probability ϕ and shuts down every firm in it. A firm's n suppliers for an input form c clusters of n/c; suppliers in a cluster share a region, and different clusters are (almost always) in different regions. With $n = 4$ we compare *spread* suppliers ($c = 4$), two *pairs* ($c = 2$), and *co-located* suppliers ($c = 1$).

Let r be the probability that a firm works given that its own region was not hit, and let $s = 1 - \phi$. A supplier whose region survived is usable with probability $p = xr$. Write $S_k(N, p)$ for the chance that at least k of N such suppliers are usable (a binomial tail). Then

$$r = F(r) := \big(\Pr[\text{an input is covered}]\big)^m, \tag{2}$$

where the coverage probability depends on the geography. With spread suppliers, each supplier survives independently, so it is $S_k(n, sxr)$. With co-located suppliers, all n survive together with probability s, so it is $s\,S_k(n, xr)$. With two pairs, both pairs survive with probability s^2 and exactly one does with probability $2s\phi$, so it is $s^2 S_k(4, xr) + 2s\phi\, S_k(2, xr)$. As in EGL, the reliability of a deep network is the largest solution r^* of (2), and the share of all firms that work is $\rho = s\,r^*$. The *tolerable shock* ϕ^* is the largest ϕ at which $r^* > 0$. For $m = 1$ and spread suppliers, (2) has the same form as DGM's equation for the $(k+1)$-core (a core vertex needs k neighbors besides the one it was reached from, as our firm needs k suppliers), so we expect sudden collapse once $k \geq 2$.

Claim 1 (Abrupt collapse)**.** *If $km \geq 2$, then as x falls or ϕ rises, reliability r^* jumps from a positive value straight to 0. If $k = m = 1$, it shrinks to 0 gradually.*

Proof. The chance that at least k of the suppliers are usable is at most $\binom{n}{k} p^k$ (add up the chances over all groups of k suppliers). So $F(r) \leq K r^{km} \leq K r^2$ with $K = \binom{n}{k}^m$. For $0 < r < 1/K$ this gives $F(r) < r$, so (2) has no solution in $(0, 1/K)$: reliability is either 0 or at least $1/K$. Since r^* moves monotonically with x and ϕ, it must jump. If $k = m = 1$, then for small r we have $F(r) \approx nsx\,r$, which is a branching process with mean nsx; its survival probability grows from 0 continuously once $nsx > 1$ [7, §21.8]. □

For $m \geq 2$ this is EGL's precipice. The new case is $m = 1$ and $k \geq 2$, where needing two suppliers of a single input is already enough for a cliff. For example, with $m = 1$, $n = 4$, $k = 2$, reliability drops from 0.745 to 0 at $x = 0.725$ (Figure 1a).

Claim 2 (Spread versus co-located suppliers)**.** *(a) With spread suppliers, a shock ϕ has exactly the effect of weakening every relationship from x to $(1-\phi)x$, so $\phi^*_{\text{spread}} = 1 - x_{\text{crit}}/x$, where x_{crit} is the no-shock critical strength for this k. (b) With one input ($m = 1$), spread and co-located suppliers have the same tolerable shock. With m inputs, $\phi^*_{\text{co}} = 1 - (1 - \phi^*_{\text{spread}})^{1/m}$, which is smaller.*

Proof. (a) With spread suppliers, $F(r) = [S_k(n, sxr)]^m$, which is EGL's recursion with strength sx; it keeps working as long as $sx \geq x_{\text{crit}}$. (b) Let $G(u) = [S_k(n, xu)]^m$. Spread suppliers give $r = G(sr)$; multiplying by s and writing $u = sr$ gives $u = sG(u)$. Co-located suppliers give $r = s^m G(r)$. So both cases lead to the same equation $u = aG(u)$, and only

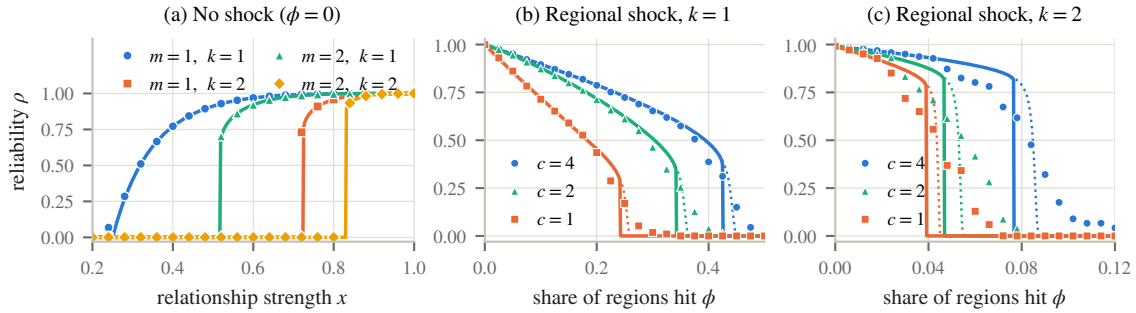


Figure 1: Simulation (markers) against theory (solid: infinite depth; dotted: 20 layers). (a) No shock, $n = 4$: $k = 2$ creates a jump even with one input (Claim 1). (b, c) Regional shocks for $m = 2$, $n = 4$, $x = 0.9$ and each number of clusters c.

the constant in front differs: $a = s$ for spread and $a = s^m$ for co-located suppliers. Raising a only raises the right-hand side, so positive solutions exist exactly when a is at least a cutoff a^* that depends only on G: spread networks collapse once $s < a^*$, and co-located ones once $s^m < a^*$. □

With several inputs, co-location gives each input its own chance of being wiped out. For $m = 2$, $n = 4$, $x = 0.9$, co-location lowers ϕ^* from 0.425 to 0.242 when $k = 1$, and from 0.077 to 0.039 when $k = 2$. Requiring $k = 2$ instead of $k = 1$ shrinks the tolerable shock more than fivefold.

Claim 3 (More clusters when $k = 1$)**.** *If $k = 1$, splitting each cluster into smaller ones (going from c to c' clusters, where c divides c') never lowers r^* or ϕ^*.*

Proof (for splitting each cluster in two; other splits work the same way). With $k = 1$, an input fails only if no surviving supplier is usable. Let $q = 1 - xr$. A cluster of b suppliers provides no usable supplier with probability $\phi + sq^b$: either it is hit, or it survives and all b fail. So $F(r) = [1 - (\phi + sq^b)^c]^m$. Splitting each cluster into two halves replaces $\phi + sq^b$ by $(\phi + sq^{b/2})^2$, and since $y \mapsto y^2$ is convex, Jensen's inequality gives $(\phi \cdot 1 + st)^2 \le \phi \cdot 1^2 + st^2$ with $t = q^{b/2}$. So each input fails less often, F goes up, and so does its largest fixed point. □

Example 1 (Pairs can be worst when $k = 2$)**.** *Claim 3 does not extend to $k \ge 2$. With $n = 4$, $k = 2$, $m = 1$, $x = 0.9$, solving* (2) *numerically gives $\phi^* = 0.195$ for co-located suppliers,* 0.171 *for two pairs, and* 0.195 *for spread suppliers.*

The two extremes tie by Claim 2(b). To see why pairs do worse, subtract the coverage formulas above: an input with two pairs is covered more often than a co-located one by $s\phi\,[2p^2 - S_2(4,p)]$, where $p = xr$. A paired firm often loses exactly one pair (probability $2s\phi$) and must then use both remaining suppliers (probability p^2), while a co-located firm is either untouched or loses all four suppliers (probability ϕ). When relationships are strong, the bracket is positive and pairs are better. Near the tipping point p is only about 0.6, below the value $2/3$ where the bracket turns negative, and pairs are worse. These barely sourced firms are like DGM's vertices with exactly k core neighbors. In a grid search over even $n \le 8$, $m \le 3$, and six values of x, such reversals appeared in 72 of the 288 settings with $k \ge 2$ and in none with $k = 1$. So for Q2, finer splits never hurt when one supplier is enough, but when firms need two, two pairs can be the worst split.

7 Simulations

Setup. The claims are about infinitely deep trees, so we check them on finite networks. A bottom layer and 20 layers above it each hold 20,000 firms, placed in 200 regions. We mostly use $n = 4$ suppliers per input (CNST's firms have about five in all) and, outside Figure 1a, $x = 0.9$, above x_{crit} for every (m,n,k) we test, so that collapse comes from the regional shock. Each firm picks its suppliers at random from the layer below, following its cluster pattern. We compare each run with the theory after 20 steps of (2), evaluated at the share of regions actually hit in that run (with only 200 regions, this share varies). Over 3,360 runs (Figure 1), simulation and 20-layer theory differ by 0.0025 on average.

Sudden collapse (Q1). We ran 300 single-input networks ($n = 4$, spread suppliers, $x = 0.9$) at the ϕ^* predicted for each k (Figure 2a). With $k = 1$, 90% of runs end with reliability below 0.1 and none above 0.21, so the network fades gradually. With $k = 2$ the runs split into two groups: 20% collapse below 0.1, while 71% stay above 0.4.

Table 1: Tolerable shock ϕ^* ($n = 4$, $x = 0.9$). Simulation: median tipping point of 40 networks [95% bootstrap CI].

m	k	Geography c	Theory (infinite depth)	Theory (20 layers)	Simulation [95% CI]
2	1	4 (spread)	0.4249	0.4437	0.4437 [0.4430, 0.4442]
2	1	1 (co-located)	0.2417	0.2543	0.2542 [0.2538, 0.2550]
2	2	4 (spread)	0.0766	0.0864	0.0867 [0.0860, 0.0875]
2	2	2 (pairs)	0.0468	0.0542	0.0542 [0.0535, 0.0545]
2	2	1 (co-located)	0.0390	0.0446	0.0447 [0.0442, 0.0450]
1	2	4 (spread)	0.1946	0.2162	0.2162 [0.2157, 0.2170]
1	2	2 (pairs)	0.1705	0.1946	0.1948 [0.1940, 0.1950]
1	2	1 (co-located)	0.1946	0.2174	0.2167 [0.2162, 0.2175]

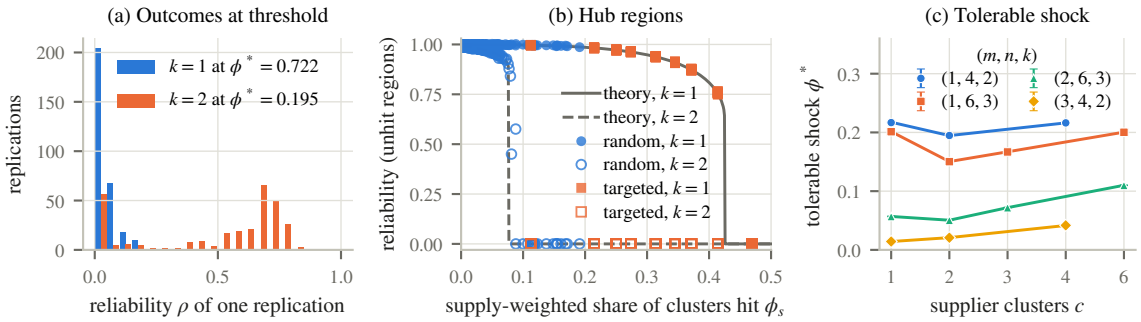


Figure 2: (a) 300 runs at the theoretical threshold ($m = 1$, $n = 4$): $k = 1$ fades, $k = 2$ splits into collapse and survival. (b) Hub regions ($m = 2$, $n = 4$, $c = 4$): runs follow the theory once the shock is measured by the share of supplier clusters hit. (c) Tolerable shock against the number of clusters c: 20-layer theory (lines), simulation (95% intervals).

Spread versus co-located (Q2). To measure tolerable shocks precisely, we use 2,000 regions of 10 firms each, hit an exact number of them, and find each network's tipping point (reliability below 0.1) by bisection (in a fixed network, hitting more regions can only hurt). In all 20 settings we simulated, the 95% bootstrap interval (widened by half a region-step) contains the 20-layer theory, at most 0.0007 from the simulated median. With two inputs, co-location lowers the tolerable shock, as Claim 2 predicts. With one input and $k = 2$, the extremes collapse together and the paired network collapses earlier than both, as in Example 1; Figure 2(c) shows a similar reversal for $n = 6$, $k = 3$.

Firms versus supply (Q3). Heavy-tailed supplier popularity (as in CNST's data) changed average reliability by at most about 0.03, since a popular supplier is no more likely to fail. Concentrating supply in a few regions mattered much more. In one setup the largest region holds 11% of supply relationships but only 0.5% of firms. Hitting that region alone collapses a $k = 2$ network, while hitting up to 10 random regions (5% of firms) collapses it in only 5% of runs. Runs match the theory (average error 0.002) when the shock is the share of supplier clusters hit (Figure 2b).

8 Further Research and Next Steps

Do firms choose fragile setups? (final project) In EGL, firms ignore how their reliability helps their customers, so the equilibrium can sit at the cliff. We will let firms also choose how many suppliers and regions to use, at a cost per supplier and per region, compute symmetric equilibria on a grid ($n \leq 6$, all c, a few costs), and compare them with a planner's choice. We expect firms to under-diversify and, when $k = 2$, sometimes to choose pairs, the worst split in Example 1.

Testing on data (final project). Our model suggests measuring a firm's exposure by the share of its suppliers *for each input* in the disaster area, not its distance to it, and predicts a sharp drop once fewer than k suppliers survive, with more exits among the most exposed. CNST's data is proprietary, so we will start with public inter-country input–output tables (OECD), treating countries as regions, and check whether industries' output losses after the 2011 Thailand floods jump once their input share from the hit region passes some level, rather than growing in proportion.

Smaller questions. When exactly are pairs worst? What changes with cycles? Can firms be protected on a budget, a reverse influence maximization [9]?

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